



4.10 Non-linear regression and the Poisson distribution (HL)

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

HL extends model fitting to further regression types (e.g. power models $y=ax^b$), still chosen using R^2 and contextual sense (as in linear/exponential fitting). The Poisson distribution $X \sim \text{Po}(m)$ models the number of random, independent events occurring in a fixed interval of time or space at a constant mean rate m : $P(X=x)=e^{-m} \frac{m^x}{x!}$... precisely $P(X=x)=e^{(-m)} m^x / x!$, with $E(X)=\text{Var}(X)=m$. It suits contexts like call arrivals, defect counts, or rare events over a fixed interval.

Worked example: follow the reasoning

A call centre receives calls at an average rate of 4 per 10-minute period, modelled by $X \sim \text{Po}(4)$. Find (a) $P(X=2)$, (b) $P(X \geq 1)$.

Solution: (a) $P(X=2)=e^{-4}(4^2)/2!=8e^{-4} \approx 0.147$. (b) $P(X \geq 1)=1-P(X=0)=1-e^{-4} \approx 1-0.0183=0.982$.

Quick check · multiple choice

For $X \sim \text{Po}(m)$, which statement is true?

- A $E(X)=m^2$
- B $E(X)=\text{Var}(X)=m$
- C $\text{Var}(X)=m^2$
- D $E(X)=2m$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A website records server errors averaging 3 per hour, modelled by $X \sim \text{Po}(3)$. (a) [7]
Find $P(X=0)$, to 3 significant figures. (b) Find $P(X \leq 2)$, to 3 significant figures. (c) A separate data set of traffic vs. loading time is fitted with a power model $y=2.5x^{0.6}$ ($R^2=0.95$) and a linear model ($R^2=0.81$) using technology. State, with reason, which model should be preferred.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $P(X=0)=e^{-3}\approx 0.0498$. (b) $P(X=1)=3e^{-3}\approx 0.149$; $P(X=2)=4.5e^{-3}\approx 0.224$;
 $P(X\leq 2)\approx 0.0498+0.149+0.224\approx 0.423$. (c) The power model should be preferred, since its $R^2=0.95$ is closer to 1 than the linear model's 0.81, meaning it explains more of the variability in the data and fits it more closely.

Quick check answer

B. Both the mean and variance of a Poisson distribution equal its parameter m .

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

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